

Learning objectives

In this lesson you will learn to:

- Define what is meant by the term 'hacker'
- Explain why unpatched software is a target for hackers
- Explain the function of a firewall
- Explain how ethical hacking and penetration testing help identify vulnerabilities.

For more information and additional learner activities see Topic 5.3 of the student book.

Hackers

What is a hacker?



Do they need good technical skills?



Can you recognise a hacker?



What motivates them?



Hackers

A hacker is a person who attempts to gain or gains unauthorised access to a computer system.

This often requires good technical knowledge, but can also be achieved by using social skills to manipulate individuals.

The motivation of hackers is harder to pin down – there are a number of reasons why people may choose to hack systems.

Motivations

Hackers may be motivated by any number of things.

- Money – stealing bank details and identity theft can be highly profitable.
- The desire to damage an organisation.
- Political motivation.
- Spying or corporate espionage.
- To help an organisation identify vulnerabilities in its systems.
- Just for fun! It's a challenge and some may just want to see if they can do it.

Easy targets

In the last lesson on malware, you learned about the WannaCry ransomware attack. This attack took advantage of a security vulnerability in MS Windows.

The organisations that were affected by WannaCry all left their software vulnerable because they had not installed the necessary software patch.

Patches are security fixes for existing software. The vulnerabilities that are fixed are often common knowledge or easily discovered. This means that unpatched software is an easy target for hackers.

How do we stop hackers?

A common tool to protect computer systems is a **firewall**.

Firewalls are software. They are designed to monitor and block incoming and outgoing connections to a network.

For large networks, the firewall software may run on a dedicated piece of hardware.

How does a firewall work?

Firewalls protect a system from hackers who are attempting to access it from outside allowable channels.

If a hacker is already inside a system by other means, they may be prevented from doing additional damage by the firewall blocking outgoing traffic.

What colour hat are they wearing?

It is important to remember that not all hackers are malicious.

Some people hack in order to help organisations protect themselves from malicious hackers. This is called **ethical hacking**.

Ethical hackers may be employed specifically to break into systems, both digitally and physically, in order to identify security weaknesses. This is called **penetration testing**.

Wrap up: you have learned how to...

- ✓ Define what is meant by the term 'hacker'.
 - A person who attempts to gain or gains unauthorised access to a computer system.
- ✓ Explain why unpatched software is a target for hackers.
 - Security vulnerabilities are often common knowledge, which means software that has not had a security patch is an easy target.
- ✓ Explain the function of a firewall.
 - A firewall monitors and blocks unauthorised incoming and outgoing traffic to a computer system.
- ✓ Explain how ethical hacking and penetration testing help identify vulnerabilities.
 - By attacking a system in the same way a 'black hat' hacker might, 'white hat' hackers can help organisations identify weaknesses in their security systems.